

COMPSYS704 – Advanced Embedded Systems Individual Research Project (IRP)

Zoran Salcic
Department of Electrical and Computer
Engineering

University of Auckland

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IP – Extended ABS and related topics

- Extension of the ABS to deal with at least two different bottle sizes, e.g. 200 ml and 500 ml (loading and unloading, filling, assume other operations are identical)
- System ability to separate completed products (full bottles) of different sizes as in the above into their own storage as part of unloading process and sorted by batches of separate products
- Inclusion of elements of fault-tolerance (e.g. mitigation of failure of some functional parts, e.g. workstation and/or transport device, possible inclusion of Baxter robot, see Appendix 5)

IP – Extended ABS and related topics

- Generalisation of the way how the recipe is presented and then interpreted by control system
- Using digital twins of products to enhance the production process and enable tracking product's after-sales life
- Any improvement on SystemJ side (run-time system)
- Other, proposed by group members/individual students
- Purchase order system (as one of the topics)
- Developing visualisation without use of Java graphics libraries and integrating it into SystemJ capabilities

IP – Research Proposal – Conceptual Design

Advice how to tackle the topic you selected:

- Discuss the topic of your choice with your group members to make decision
- Narrow down the topic and make additional literature search as well as Internet search if applicable and necessary
- Make short proposal and conceptual design first to discuss with teaching team and get a “green light” to continue



IP – Research Proposal

Write project proposal (conceptual design) for the milestone 1 submission:

- Define narrower topic and problem you are going to explore/solve
- Think of what is challenging in your topic and where there is potential to make contributions (research / development); list them and elaborate in a sentence or two – leave some things for the discussion with teaching staff
- Think out of box - modify the proposal to suit your thinking, background knowledge – don't be constrained by anything; discuss with lecturers if in doubt
- Use an example as the driving / motivating idea
- Make a diagram/sketch of the conceptual design



IP – Research Proposal

How it looks like and additional advice:

- Minimum 2 A4 single-spaced pages (min 600 words including additional figures /tables / diagrams if you choose)
- References do not count in the word count!
- Submit via Canvas

IP – Final report and presentation

How it looks like and additional advice:

- The report must be minimum four A4 pages long, font 10pt, single spaced, with additional diagrams and figures if necessary
- Additional compendium with SystemJ and Java codes (project) of implemented system/solution
- Submit via Canvas
- Presentation, PPT with four slides (plus cover/title slide)